

# Lecture 16: Difference-in-differences

## Economics 326 — Introduction to Econometrics II

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### Motivation

- In the previous lecture, we estimated treatment effects from **cross-sectional** data under the assumption of selection on observables: after controlling for covariates, treatment is as good as random.
- In many settings, this assumption is hard to justify. An alternative approach exploits **panel data** (repeated observations on the same units over time).
- The **difference-in-differences (DID)** method compares changes over time between a treatment group and a control group.

### DID basic setup

- Two time periods:  $t \in \{0, 1\}$  (before and after treatment).
- Two groups:  $D_i \in \{0, 1\}$  (control and treatment).
- Treatment occurs between periods 0 and 1, and only the treatment group ( $D_i = 1$ ) is affected.
- We observe  $Y_{it}$ : the outcome for individual  $i$  at time  $t$ .

### DID regression model

- The DID regression is:

$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it},$$

where  $E[U_{it} \mid D_i] = 0$ .

- The regressors:
  - $t$ : time indicator (0 = before, 1 = after).
  - $D_i$ : group indicator (0 = control, 1 = treatment).
  - $t \cdot D_i$ : interaction term, equals 1 only for the treatment group after treatment.

### Interpreting the coefficients

•

$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it},$$

- Evaluate  $E[Y_{it} \mid D_i]$  for each combination of  $t$  and  $D_i$ :

$$t = 0, D_i = 0: E[Y_{i0} \mid D_i = 0] = \alpha,$$

$$t = 0, D_i = 1: E[Y_{i0} \mid D_i = 1] = \alpha + \gamma,$$

$$t = 1, D_i = 0: E[Y_{i1} \mid D_i = 0] = \alpha + \delta,$$

$$t = 1, D_i = 1: E[Y_{i1} \mid D_i = 1] = \alpha + \delta + \gamma + \beta.$$

- Summarized as a 2×2 table:

|         | $D_i = 0$ (Control) | $D_i = 1$ (Treatment)              |
|---------|---------------------|------------------------------------|
| $t = 0$ | $\alpha$            | $\alpha + \gamma$                  |
| $t = 1$ | $\alpha + \delta$   | $\alpha + \delta + \gamma + \beta$ |

- $\alpha$ : baseline expected outcome (control group, before treatment).
- $\gamma$ : pre-existing **group difference** at baseline ( $t = 0$ ).
- $\delta$ : **time effect** — the change in the control group from  $t = 0$  to  $t = 1$ , capturing common trends.
- From the 2×2 table, the change over time for each group:
  - Treatment:**  $E[Y_{i1} | D_i = 1] - E[Y_{i0} | D_i = 1] = \delta + \beta$ .
  - Control:**  $E[Y_{i1} | D_i = 0] - E[Y_{i0} | D_i = 0] = \delta$ .
- Subtracting the control group's change from the treatment group's change:

$$(\delta + \beta) - \delta = \beta.$$

- The DID estimand as a double difference of conditional expectations:

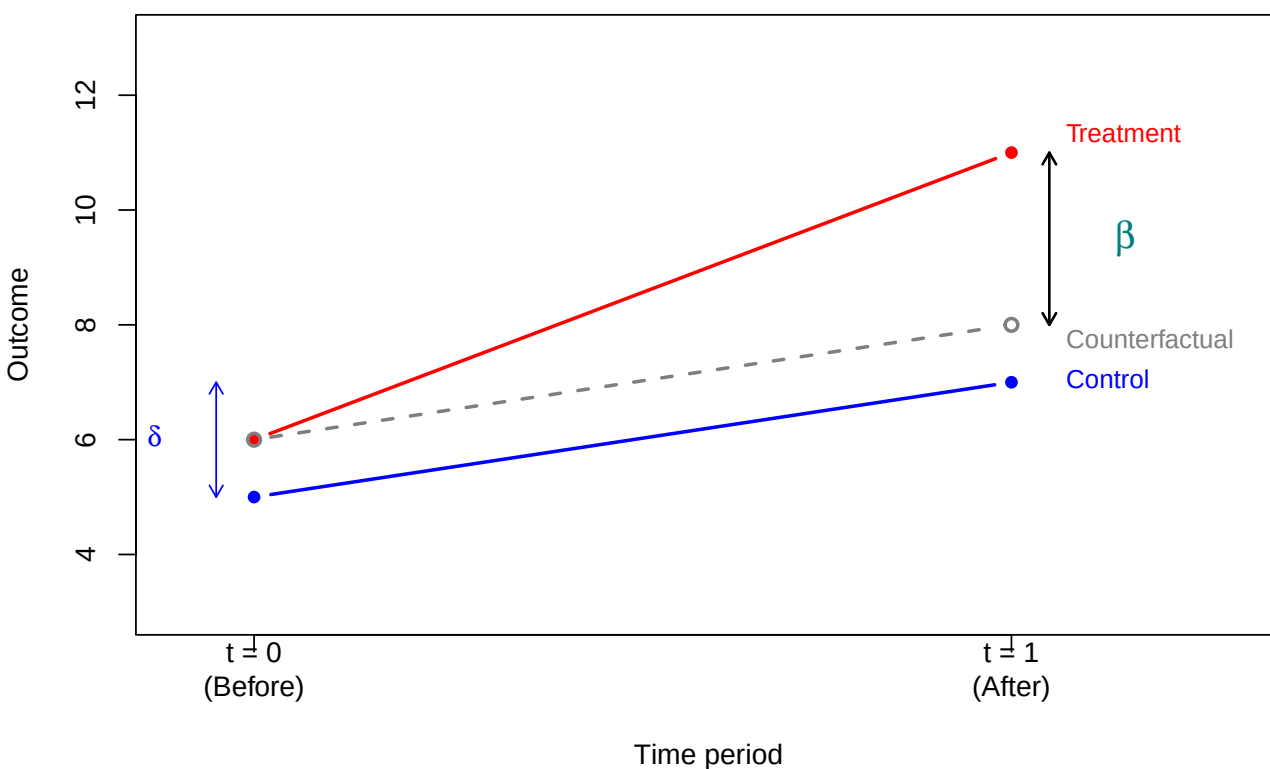
$$\beta = E[Y_{i1} - Y_{i0} | D_i = 1] - E[Y_{i1} - Y_{i0} | D_i = 0].$$

- By subtracting the control group's change, the common time trend  $\delta$  cancels, isolating the treatment effect  $\beta$ .

## DID diagram

- The classic DID diagram shows the control group, treatment group, and counterfactual:

### Difference-in-differences



- We predict the counterfactual outcome for the treatment group at  $t = 1$  (dashed gray line) by adding the control group's change  $\delta$  to the treatment group's baseline  $\alpha + \gamma$ .

## DID and potential outcomes

- To connect DID with the potential outcomes framework, define **panel potential outcomes**:  $Y_{it}(d)$  is the outcome for individual  $i$  at time  $t$  if assigned to group  $d \in \{0, 1\}$ .
- Four potential outcomes:  $Y_{i0}(0)$ ,  $Y_{i1}(0)$ ,  $Y_{i0}(1)$ ,  $Y_{i1}(1)$ .
- The observed outcome is:

$$Y_{it} = D_i Y_{it}(1) + (1 - D_i) Y_{it}(0).$$

- What we observe for each group:

|         | Control ( $D_i = 0$ ) | Treatment ( $D_i = 1$ ) |
|---------|-----------------------|-------------------------|
| $t = 0$ | $Y_{i0}(0)$           | $Y_{i0}(1)$             |
| $t = 1$ | $Y_{i1}(0)$           | $Y_{i1}(1)$             |

- The treatment effect at time  $t = 1$  for the treated group is:

$$ATT = E[Y_{i1}(1) - Y_{i1}(0) | D_i = 1].$$

The counterfactual  $Y_{i1}(0)$  is unobserved for the treated group.

## DID as a treatment effect

- $\beta = E[Y_{i1} - Y_{i0} | D_i = 1] - E[Y_{i1} - Y_{i0} | D_i = 0]$ .
- Substituting observed outcomes with potential outcomes:

$$\beta = E[Y_{i1}(1) - Y_{i0}(1) | D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) | D_i = 0].$$

- To relate  $\beta$  to the ATT, add and subtract  $E[Y_{i1}(0) | D_i = 1]$  and  $E[Y_{i0}(0) | D_i = 1]$  inside the first expectation:

$$\begin{aligned} \beta &= E[Y_{i1}(1) - Y_{i0}(1) | D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) | D_i = 0] \\ &= E\left[Y_{i1}(1) - \underbrace{Y_{i1}(0) + Y_{i1}(0)}_{=0} - \underbrace{Y_{i0}(0) + Y_{i0}(0)}_{=0} - Y_{i0}(1) | D_i = 1\right] \\ &\quad - E[Y_{i1}(0) - Y_{i0}(0) | D_i = 0]. \end{aligned}$$

- Rearranging and splitting the expectation:

$$\begin{aligned} \beta &= \underbrace{E[Y_{i1}(1) - Y_{i1}(0) | D_i = 1]}_{ATT} \\ &\quad + \underbrace{E[Y_{i1}(0) - Y_{i0}(0) | D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) | D_i = 0]}_{\text{difference in trends}} \\ &\quad + \underbrace{E[Y_{i0}(0) - Y_{i0}(1) | D_i = 1]}_{\text{anticipation effect}}. \end{aligned}$$

- For  $\beta$  to equal the **ATT**, the **difference in trends** and **anticipation effect** must be zero. This requires two assumptions.

### Assumption 1: Parallel trends

- The “difference in trends” term equals zero if both groups would have experienced the same change over time in the absence of treatment:

$$E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] = E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0].$$

- Under parallel trends, the decomposition reduces to:

$$\beta = \text{ATT} + \underbrace{E[Y_{i0}(0) - Y_{i0}(1) \mid D_i = 1]}_{\text{anticipation effect}}.$$

- The parallel trends assumption cannot be directly tested because  $Y_{i1}(0)$  is unobserved for the treated group.
- If pre-treatment data for multiple periods exist, one can check whether trends were parallel before treatment.

### Assumption 2: No anticipation

- The “anticipation effect” term equals zero if the outcome at  $t = 0$  (before treatment) is not affected by future treatment assignment:

$$E[Y_{i0}(1) \mid D_i = 1] = E[Y_{i0}(0) \mid D_i = 1].$$

- Being assigned to the treatment group does not change pre-treatment outcomes in expectation.
- Under both parallel trends and no anticipation:

$$\beta = \text{ATT}.$$

### Example: incinerator and house prices

- Kiel and McClain (1995) studied how the construction of a garbage incinerator affected nearby house prices in North Andover, Massachusetts. This is an example of an **event study**: a research design that estimates the causal effect of a specific event by comparing outcomes before and after it occurs.
- We use the `kielmc` dataset from the `wooldridge` package:

```
library(wooldridge)
data(kielmc)
# Show 2 observations from each of the 4 groups
rows <- c(
  head(which(kielmc$y81 == 0 & kielmc$nearinc == 0), 2),
  head(which(kielmc$y81 == 0 & kielmc$nearinc == 1), 2),
  head(which(kielmc$y81 == 1 & kielmc$nearinc == 0), 2),
  head(which(kielmc$y81 == 1 & kielmc$nearinc == 1), 2)
)
kielmc[rows, c("rprice", "y81", "nearinc", "age")]
```

|     | rprice   | y81 | nearinc | age |
|-----|----------|-----|---------|-----|
| 14  | 52000.00 | 0   | 0       | 32  |
| 15  | 49000.00 | 0   | 0       | 18  |
| 1   | 60000.00 | 0   | 1       | 48  |
| 2   | 40000.00 | 0   | 1       | 83  |
| 187 | 90245.77 | 1   | 0       | 1   |
| 188 | 46082.95 | 1   | 0       | 41  |
| 180 | 37634.41 | 1   | 1       | 81  |
| 181 | 39938.55 | 1   | 1       | 71  |

- **rprice**: house price in 1978 dollars ( $Y_{it}$ ).
- **y81**: 1 if year is 1981 (after incinerator announced), 0 if 1978 ( $t = 1$  if 1981, 0 if 1978).
- **nearinc**: 1 if house is near the incinerator site ( $D_i$ ).

## The 2×2 table of means

- Compute the four group means:

```
means <- tapply(kielmc$rprice, list(kielmc$y81, kielmc$nearinc), mean)
colnames(means) <- c("Far (nearinc=0)", "Near (nearinc=1)")
rownames(means) <- c("1978 (y81=0)", "1981 (y81=1)")
round(means, 2)
```

|              | Far (nearinc=0) | Near (nearinc=1) |
|--------------|-----------------|------------------|
| 1978 (y81=0) | 82517.23        | 63692.86         |
| 1981 (y81=1) | 101307.51       | 70619.24         |

- Computing the DID by hand:

```
diff_near <- means[2, 2] - means[1, 2]
diff_far <- means[2, 1] - means[1, 1]
DID <- diff_near - diff_far
cat("Change (near):", round(diff_near, 2), "\n")
```

Change (near): 6926.38

```
cat("Change (far): ", round(diff_far, 2), "\n")
```

Change (far): 18790.29

```
cat("DID: ", round(DID, 2), "\n")
```

DID: -11863.9

## DID regression

- The DID regression, where  $y81nrinc = y81 \times nearinc$  is the interaction term:

```
options(scipen = 999)
reg_did <- lm(rprice ~ y81 + nearinc + y81nrinc, data = kielmc)
round(summary(reg_did)$coefficients, 4)
```

|             | Estimate  | Std. Error | t value | Pr(> t ) |
|-------------|-----------|------------|---------|----------|
| (Intercept) | 82517.23  | 2726.910   | 30.2603 | 0.0000   |
| y81         | 18790.29  | 4050.065   | 4.6395  | 0.0000   |
| nearinc     | -18824.37 | 4875.322   | -3.8612 | 0.0001   |
| y81nrinc    | -11863.90 | 7456.646   | -1.5911 | 0.1126   |

- The coefficient on  $y81nrinc$  matches the DID computed from the 2×2 table.
- The estimated DID effect is  $\hat{\beta} = -\$11,864$  (SE = 7,457,  $p = 0.113$ ). The incinerator reduced nearby house prices, but the effect is not statistically significant at the 5% level.

## Assumptions in the incinerator example

- **Parallel trends:** without the incinerator, house prices near and far from the site would have followed the same trend over time.
- **No anticipation:** before the incinerator was announced, living near the future site did not affect house prices.

## DID with covariates

- The basic DID estimator is unbiased only if the parallel trends assumption holds unconditionally. If houses near the incinerator site are systematically different from those farther away (e.g., older), and house age affects price trends, then near and far houses may follow different price trajectories even without the incinerator. This violates unconditional parallel trends.
- Omitting age from the regression biases the estimated common trend  $\delta$ , which in turn biases the DID estimate  $\beta$ .
- Adding covariates addresses this under a weaker assumption: **parallel trends holds conditional on house age**. Among houses of the same age, near and far houses would have followed the same price trend absent the incinerator.

```
reg_did_cov <- lm(rprice ~ y81 + nearinc + y81nrinc + age + I(age^2),
                  data = kielmc)
round(summary(reg_did_cov)$coefficients, 4)
```

|             | Estimate    | Std. Error | t value  | Pr(> t ) |
|-------------|-------------|------------|----------|----------|
| (Intercept) | 89116.5354  | 2406.0511  | 37.0385  | 0.0000   |
| y81         | 21321.0418  | 3443.6311  | 6.1914   | 0.0000   |
| nearinc     | 9397.9359   | 4812.2218  | 1.9529   | 0.0517   |
| y81nrinc    | -21920.2700 | 6359.7454  | -3.4467  | 0.0006   |
| age         | -1494.4240  | 131.8603   | -11.3334 | 0.0000   |
| I(age^2)    | 8.6913      | 0.8481     | 10.2476  | 0.0000   |

- After controlling for house age, the DID estimate is  $\hat{\beta} = -\$21,920$  (SE = 6,360,  $p = 0.0006$ ) — nearly twice as large and highly significant.
- The change in the estimate suggests that the basic DID was biased: omitting age biased the estimated trend, which in turn biased  $\hat{\beta}$ .

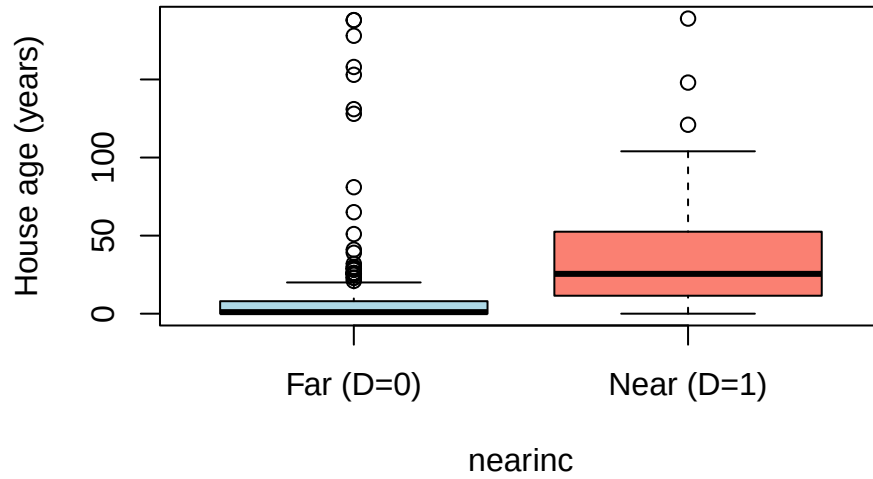
## Where does the bias come from?

- Under no anticipation, the decomposition from the earlier slide gives:

$$\beta = \text{ATT} + \underbrace{E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]}_{\text{difference in trends}}.$$

- **No anticipation is plausible:** the incinerator was not publicly known in 1978, so there is no reason for 1978 prices to already reflect the future incinerator.
- The bias must come from the **difference in trends**: a violation of **parallel trends**.
- Houses near the incinerator are systematically older:

```
boxplot(age ~ nearinc, data = kielmc,
        names = c("Far (D=0)", "Near (D=1)"),
        ylab = "House age (years)",
        col = c("lightblue", "salmon"))
```



- Since house age affects price trends nonlinearly, the near and far groups would have had **different price trends even without the incinerator**.
- The estimated bias is approximately  $-\$11,864 - (-\$21,920) \approx +\$10,056$ : the age-driven trend difference partially masked the incinerator's negative effect.

| Assumption      | Violated?             | Reasoning                                                      |
|-----------------|-----------------------|----------------------------------------------------------------|
| No anticipation | No                    | Incinerator not yet announced in 1978                          |
| Parallel trends | Yes (unconditionally) | Near houses are older; older houses had different price trends |

## Detecting anticipation and pre-trends

- With only two time periods (as in the incinerator example), anticipation effects **cannot** be separately identified.
- With **multiple pre-treatment periods**, an **event study** design can help. Include leads and lags of the treatment indicator:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq 0} \mu_k \cdot D_i \cdot \mathbb{1}[t = k] + U_{it},$$

where  $k < 0$  are pre-treatment periods and  $k > 0$  are post-treatment periods.

- The pre-treatment coefficients  $\mu_k$  for  $k < 0$  serve as a diagnostic:

| Pattern in $\mu_k$ for $k < 0$                                                                  | Interpretation                                                                                      |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| All close to zero: $\mu_{-3} \approx \mu_{-2} \approx \mu_{-1} \approx 0$                       | Both assumptions look plausible                                                                     |
| Sharp break right before treatment: $\mu_{-3} \approx \mu_{-2} \approx 0$ but $\mu_{-1} \neq 0$ | <b>Anticipation:</b> units react to news of upcoming treatment                                      |
| Gradual trend across pre-periods: $ \mu_{-3}  <  \mu_{-2}  <  \mu_{-1} $ , growing steadily     | <b>Parallel trends violation</b> , not anticipation: groups were already diverging before treatment |